



Semester -III

BRANCH	SubjectTitle	SubjectCode
B.Sc.(Zoology–Minor)	Physiology and Biochemistry	

Course Objective: -

- Students would be able to know basic fundamentals and understand advanced concepts related to systems in the body, their feedback loop controls.
- They would be able to know the role of self-sustaining systems like circulatory, digestive, respiratory and excretory systems and how all of these work in unison to maintain a balance in the body.
- understand the processes in metabolism and regulation of metabolic pathways.
- understand the metabolism of carbohydrates, proteins and lipids through various anabolic and catabolic pathways like glycolysis, gluconeogenesis, Krebs cycle, Glycogen metabolism, transamination, deamination, urea cycle, beta and omega oxidation of saturated fatty acids and their regulation; Ketogenesis.
- know in detail about concepts to illustrate how enzymes and redox carriers and the oxidative phosphorylation machinery occur.
- Get exposure to various processes of metabolism and quantification of many biomolecules.
- Plan and safely perform biochemical and physiological significant enzyme assays.

Course Learning outcomes(CLO)

1. Upon completion of the course, Students will be able to 1 Understand how organs function at different levels i.e. from cellular to system levels.
2. Examine internal harmony of different body systems by learning inherent disorders and deficiencies, which is needed to maintain good health.
3. Understand functions of biomolecules & their role in metabolism by studying biochemistry.
4. Develop a strong foundation for research & employability skills
5. Improve the student's perspective of health biology through deep study of physiology..



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Unit -I

Introduction and Historical background of Physiology and Biochemistry Biomolecules and Regulatory mechanism.

1. Contribution of Indian Scientists

- 1.1 Contribution of Charak
- 1.2 Contribution of Sushrut

2. Biomolecules

- 2.1 Micro and Macro molecules
- 2.2 Water and Buffer System

3. Enzymes

- 3.1 Definition and General Properties
- 3.2 Nomenclature and Classification and functions
- 3.3 Mechanism and Regulation of Enzyme action
- 3.4 Co-Enzyme

4. Vitamins and Minerals

- 4.1 Types and Sources
- 4.2 Biological importance
- 4.3 Deficiencies and Disorders



Unit-II

Metabolism, Physiology and Regulation

1. Protein

- 1.1 Structure, Nomenclature, Classification and Biological importance.
- 1.2 Metabolism -Deamination, Decarboxylation, Transamination of amino acids and Ornithine cycle

2. Carbohydrates

- 2.1 Structure, Nomenclature, Classification and Biological importance.
- 2.2 Metabolism -Glycogenesis, Gluconeogenesis, Glycogenolysis, Glycolysis. Citric Acid Cycle and Electron Transport Chain

3. Lipids

- 3.1 Structure, Classification and Biological importance
- 3.2 Metabolism -Beta oxidation of fatty acids.

4. Physiology of Digestion, regulation and disorders

5. Homeostasis and Basal Metabolic rate (BMR)

6. Thermoregulation

Unit - III

Respiration, Excretion and Immune System

1. Respiration

- 1.1 Mechanism -Inspiration and Expiration
- 1.2 Physiology- Exchange and Transport of Gases (Oxygen and carbon dioxide), Chloride shift, role of Respiratory pigment.
- 1.3 Disorders - Apnea, Hypoxia, Asphyxia, Carbon monoxide poisoning. Bronchitis, Asthma

2. Excretion

- 2.1 Physiology -Urea, Urine formation and Counter Current mechanism
- 2.2 Excretory products, disorders
- 2.3 Osmoregulation

3. Immunity

- 3.1 Innate and acquired Immunity
- 3.2 Immune cells and ImmunoGobulinus



3.3 Antigen responses

Unit-IV

Neuromuscular Co-ordination

1. Nerves

1.1 Structure and type of Neurons

1.2 Physiology of nerve impulse conduction

1.3 Neuromuscular disorders -Epilepsy, Alzheimer's and Parkinson's disease

2. Muscles

2.1 Structure and type of muscles

2.2 Physiology of muscles contraction and its Biochemistry

2.3 Muscular disorders - Fatigue

Unit- V

Hormones, Endocrine system and Reproductive Physiology

1. Hormones

1.1 Definition and Classification

1.2 Mechanism of hormone action

2. Endocrine system

2.1 Structure, functions and disorders of Pituitary gland

2.2 Structure, functions and disorders of Thyroid and Parathyroid gland

2.3 Structure, functions and disorders of Adrenal gland

2.4 Structure, functions and disorders of Thymus gland, Pineal gland and Pancreas

3. Reproductive Physiology

3.1 Physiology of reproduction

3.2 Sex Hormones



Practical

1. Qualitative estimations of Protein, Carbohydrates and Lipids.
2. Study of effect of temperature and pH on salivary amylase activity.
3. Study of enzymatic activity of Trypsin and Lipase.
4. . Detection of ammonia, urea and uric acid
5. Estimation of hemoglobin using haemometer.
6. Preparation of haemin crystals.
7. . Preparation of blood smear, study and identification of blood cells.
8. Determination of ABO blood groups. RBC, WBC counting
9. Measurement of blood pressure using sphygmomanometer.
10. Principles and uses of instruments-Sphygmomanometer, Stethoscope,iochemistry analyzer
11. Study of endocrine glands through histological slides of pituitary gland, adrenal gland, thyroid gland, pancreas, testis, ovary, spleen and thymus.
12. Study of histological slides of organ systems of mammalian oesophagus, stomach, duodenum, ileum, rectum, liver, trachea, lung, and kidney.



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Semester– IV

BRANCH	SubjectTitle	SubjectCode
B.Sc.(Zoology–Minor)	Diversity of Chordates and Comparative Anatomy	

Course Objective: - Zoology is one of the most fundamental branch of biology to be studied at undergraduate level. It is required to learn and understand about animal diversity to appreciate the variability in relation to their morphology, anatomy and behaviour among different animals. Zoology course also provides a knowledge of applied subjects to develop various skills to make a career and become an entrepreneur in the field of aquatic biology, sericulture, apiculture etc. After completion of this course students will be able to contribute as policy makers in wild life conservation animal preservation and environment protection.

- Understand the evolutionary history and relationship between the different classes of chordates.
- Know the different characteristics along with their habits, habitats and distribution of the chordates.
- Understand the significance of the differences in physiological systems between the vertebrates.
- Distinguish the significance of the chordates from other lower organisms and comprehend their advantages.
- Understand the pattern of vertebrate evolution, organisation and functions of various systems.
- Learn the comparative account of integument, skeletal components, their functions and modifications in different vertebrates.
- Understand the evolution of heart, modification in aortic arches, structure of respiratory organs used in aquatic, terrestrial and aerial vertebrates; and digestive system and its anatomical specializations with respect to different diets and feeding habits.
- Learn the evolution of brain, sense organs and excretory organs to a complex, highly evolved form in mammals
- Learn to analyze and critically evaluate the structure and functions of vertebrate systems, which helps them to discern the developmental, functional and evolutionary history of vertebrate species.

Course learning outcomes:-

1. Understand chordate diversity of animals and their taxonomic position.
2. Identify the morphological and anatomical features and basis of chordate classification
3. Know economic importance and present status that will develop positive attitude towards conservation of biodiversity.
4. Differentiate the organism belonging to different taxa by studying comparative anatomy.
5. The project, assignment will give them a flavor of research in studying biodiversity, taxonomy besides improving their writing skills and lay foundation of career in Zoology.



Unit-I

1. Introduction to Chordates

- 1.1 Traditional Knowledge on Animal Science in ancient Indian Civilization
- 1.2 Origin of Chordates, General characteristics and outline classification of Phylum Chordata up to orders

2. Protochordata

- 2.1. General characteristics and classification of Sub- Phylum Urochordata and Cephalochordata.
- 2.2. Type study of Herdmania and retrogressive metamorphosis in ascidian tadpole.
- 2.3. Type study of Amphioxus and its Affinities.

3. Agnatha

- 3.1 Comparison of Petromyzon and Myxine.

Unit - 2

1. Pisces:

- 1.1. General characteristics and classification of Pisces.
- 1.2. Accessory respiratory organs, Parental care in fishes.

2. Amphibia

- 2.1 General characteristics and classification of Amphibia.
- 2.2 Parental Care in Amphibia and Paedomorphosis

3. Reptilia

- 3.1. General Characteristics and classification of Reptilia.
- 3.2. Difference between Poisonous and Non Poisonous snakes, Venom and Antivenom
- 3.3. Poison apparatus and biting mechanism in snake.

Unit - III

1. Aves

- 1.1. Brief Introduction of "Birdman" of India - Dr. Salim Ali
- 1.2. General characteristics and classification of Aves.
- 1.3. Migration of birds, principles and aerodynamics of flight.
- 1.4. Flight adaptation in birds.

2. Mammalia

- 2.1. General characteristics and classification of mammals.
- 2.2. Adaptive radiation in mammals with reference to locomotory appendages.
- 2.3. Introduction of ZSI (Zoological Survey of India)

Unit - IV

Comparative Anatomy of Vertebrates

- 1. Comparative study of integument and its derivatives of Vertebrates.
- 2. Comparative study of appendicular skeleton (Limb and girdles) of Vertebrates.
- 3 Comparative study of digestive system of Vertebrates.
- 4. Comparative study of respiratory system of Vertebrates.



Unit - V

Comparative Anatomy of Vertebrates

1. Comparative study of aortic arches and heart of Vertebrates.
2. Comparative study of Brain of Vertebrates.
3. Comparative study of Urinogenital System of Vertebrates
4. Study of Eye and Ear of mammals

PRACTICALS

1: Study of museum specimens

1. Protochordata: Herdmania, Amphioxus
2. Fishes: Scoliodon, Stegostoma, Torpedo, Heteropneustes, Labeo, Exocoetus, Hippocampus, Anabas, Eel, Flat fish.
3. Amphibia: Necturus, Bufo, Rana, Salamander, Hyla, Axolotl larva, Midwife Toad, Ichthyophis
4. Reptilia: Chelone, Trionyx, Hemidactylus, Varanus, Chameleon, Draco, Viper, Naja, Hydrophis.
5. Aves: Local Birds, Vulture, Great Indian Bustard, Lesser Frigatebird, Mammalia: Bat, Funambulus, Platypus, Rat,

2. Study of Histological slides-

T.S. of Duodenum, Stomach, Small Intestine, Liver, Pancreas, Testis, Ovary, V.S. of skin, L.S. of Kidney of vertebrates

3. Osteology

Study of Limb Bones and Girdles of Vertebrates (Amphibia, Reptilia, Aves, Mammalia).

4. Study of different types of feathers/ beaks of birds.

5. Dissection of Local fish (Only demonstration of commercially available local fish / Through computer simulation method/through YouTube videos / through models and charts.

a) General Viscera, arterial system

b) Cranial nerves V, VII, IX and X

6 . Mounting of scales of fishes

7 . Comparative study of heart and brain of vertebrates

8. Study of local bird fauna of surrounding area (College campus/ Village Garden/ Ward)